

Health Assistants for Everyday Care: Reliable Human-AI Collaboration under Imperfect Sensing

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Abstract

Artificial intelligence (AI) in healthcare has excelled in clinical tasks such as diagnostics and patient monitoring. Yet, it still falls short in meeting the daily needs of patients with chronic conditions and cognitive impairments, particularly in providing in-situ assistance for everyday care. My research aims to develop intelligent health assistants that go beyond reactive alerts and passive monitoring to provide proactive, context-aware interventions to support it. Key contributions include advancing sensing technologies to capture richer, real-world contextual data and designing assistive systems that support both patients and caregivers and adapt through daily interactions. I evaluate these systems in diverse domains, including post-operative skin cancer care, dementia support, and child hyperactivity management. Through these field deployments, my thesis aims to articulate design strategies for effective human-AI collaboration in the presence of sensing uncertainty, ultimately promoting positive behavior change and fostering greater autonomy in everyday care.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; *Interactive systems and tools*.

Keywords

health assistants, ubiquitous sensing, human-AI collaboration

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1 Problem Statement

AI for health and well-being research is at the forefront of tackling important issues within healthcare systems worldwide, such as misdiagnosis, mental health, and inequity. Despite significant advancements, most AI applications remain concentrated in clinical settings, where they support high-stakes monitoring and decision-making [28]. However, individuals managing chronic conditions or cognitive impairments, such as dementia, require continuous care support embedded seamlessly into their everyday routines to

prevent errors and ensure safety. Effective management of these conditions demands contextually aware assistance that promotes meaningful behavioral adaptation [31]. My vision is for everyone to have a personal health assistant integrated into daily life, enhancing individual health and independence in a scalable manner.

Such assistants must sense user behavior non-intrusively in their daily lives, interpret it accurately, and proactively intervene or offer support appropriately. With the growing proliferation of mobiles and wearables, sensing our everyday behaviors, such as activity, body pose, and speech, will be foundational for such applications. However, natural user behavior is inherently complex, and sensing systems often struggle with imperfect and uncertain data. First, ubiquitous sensing technology faces a trade-off between sensing richness and practicality. Camera-based systems offer detailed observations but raise privacy concerns, while more comfortable and privacy-preserving options like wearables provide limited sensing capabilities [1]. This limitation can lead to incorrect interactions and potentially harm users. Even with accurate sensing models, real-world deployment introduces further complexity due to unexpected user behavior and environmental noise. For example, a hand-washing detection algorithm on a commercial smartwatch can be easily triggered by similar activities, such as brushing teeth. Such false positives or negatives can reduce user trust and long-term adoption of the technology [25]. These are not just sensing failures; they are fundamental challenges in building robust human-AI systems in our physical space. To overcome them, we must design effective *Human-AI Collaboration*, where the sensing-powered assistant and user work together toward the shared goal.

2 Related Work

In this section, I focus on two key threads of work that are most relevant to the scope of this thesis: (1) technological support for chronic conditions in everyday settings, and (2) sensing technologies using common devices for behavior recognition.

2.1 Technological Support for Chronic Conditions

Researchers explore a variety of technologies supporting people with chronic conditions such as dementia, ADHD/hyperactivity, and other long-term health needs. Mobile health (mHealth) apps are widespread in dementia care, offering features like GPS-based location tracking for wander management, medication reminders, caregiver communication, and shared care planning with providers [2]. Such apps and assistive tools aim to maintain daily functioning and safety for those with cognitive impairments. In parallel, wearable



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devices and specialized interventions have shown promise for conditions like ADHD. For example, the Revibe Connect smartwatch provides adaptive vibrotactile prompts in real time to help children with ADHD self-monitor and sustain attention in class [21]. Moreover, wearable activity trackers and connected health devices (e.g., continuous glucose monitors for diabetes or smart blood pressure cuffs) enable patients to monitor symptoms and lifestyle metrics in real time [22]. These tools are promising to improve adherence to exercise, diet, and medication regimens, thereby empowering patients in day-to-day self-management.

Despite their benefits, today's mainstream health technologies often suffer critical shortcomings in adaptability. Most mHealth apps are not context-aware; they deliver generic reminders or interventions on fixed schedules, following a one-size-fits-all logic that ignores the user's current situation. These approaches fail to accommodate the dynamic, multifaceted nature of chronic conditions. In the ADHD context, for instance, traditional cognitive training games showed limited real-world transfer, in part because they were divorced from the child's day-to-day context (e.g., practiced at home instead of in the classroom). These limitations underscore the importance of designing systems that are not only present in everyday life but also sensitive to the nuances of real-time behavior and environmental context, goals central to my thesis.

2.2 Behavior Sensing Technologies with Common Devices

Ubicomp and HCI research communities have explored various ways to enable rich sensing with common devices. Human activity recognition (HAR) has been widely studied, supporting classification of movement patterns (e.g., walking, cooking, or sleeping) in both lab and home environments. Researchers look at different modalities and devices, including microphones [26], IMU sensors [24], a combination of these [29], or a machine-knitted, wearable scarf [35]. In parallel, human body pose estimation using sparse wearable sensors has gained attention; for example, using IMU-equipped devices like a watch [30, 36], phone [34], or glasses [20]. These sensing technologies are promising to capture user behavior in daily lives to offer a variety of health-related support.

However, bringing these systems into real-world health contexts remains challenging. Sensor data collected in natural settings is often noisy, incomplete, and context-ambiguous, reducing the reliability of AI models trained in controlled conditions. Additionally, systems often fail to adapt to behavioral variability across users or evolving routines, which is crucial for chronic care. These limitations highlight the need for a new class of sensing systems that go beyond passive observation and adapt to each user. My thesis argues for collaborative, co-adaptive systems that not only recognize user context but engage users in handling uncertainty and building long-term, trustworthy interactions, a vision that bridges technical robustness and human-centered design.

3 Methodology

To address the issue of imperfect sensing for health assistants, I combine sensing data with user input through daily interactions. This enables the assistant to recognize uncertainty in sensing and seek clarification when needed, while also allowing users to proactively

assist the system to improve support. This creates a reciprocal relationship in which human and AI collaborate—each strengthening the other's capabilities [27]. In my thesis, I evaluate this approach through deployments in real-world clinical contexts, including with post-operative skin cancer patients, individuals with dementia, and children with ADHD/hyperactivity, all of whom suffer from chronic or persistent conditions and therefore require situated, continuous support in their everyday lives. These projects contribute core innovations in ubiquitous sensing approaches, methodologies for enabling Human-AI Collaboration, especially using imperfect sensing, and insights from real-world deployment.

3.1 PrISM: Context-Aware Chatbots for Patients

Every day, we perform numerous tasks—ranging from cooking to medical self-care—that consist of a series of atomic steps. Accurately completing these steps can be challenging, especially when tasks become routine, fail to capture our full attention, or when individuals face cognitive impairments such as dementia. Errors like omitting essential steps or performing them in the wrong order can lead to adverse outcomes, particularly in medical contexts. This project, **Procedural Interaction from Sensing Module (PrISM)**, aims to support users in executing these tasks through dialogue-based interactions. By using multimodal sensors such as wearable devices like a smartwatch and privacy-preserving ambient sensors like a Doppler Radar, an assistant can infer the user's context and provide contextually situated help. I explore the effectiveness of real-time human-AI collaboration in fostering user independence and facilitating skill acquisition.

Figure 1 presents the PrISM framework, which is designed to be modular and flexible, allowing the integration of various sensors and interactions for each specific situation. The underlying technology is multimodal HAR and step tracking, which allow the assistant to understand the user's current step within the task. A variety of sensors can be used, while I currently focus on a smartwatch for its ubiquity, minimal privacy concerns compared to camera-based systems, and capability for monitoring a user across various daily activities. Then, the assistant offers two types of interactions: question answering and proactive intervention. In Q&A, the assistant answers the user's questions (e.g., "*What should I do next?*"), where Large Language Model (LLM)s are complemented by step context to generate more accurate responses. In Observer, the assistant proactively intervenes when it detects that the user is about to make mistakes (e.g., "*Have you wiped the pan?*") by probabilistically modeling and forecasting user behavior using HAR.

The PrISM assistant is developed to help patients who suffer from chronic conditions. For instance, patients recovering from Mohs surgery are at ongoing risk of infection due to mistakes during self-care procedures [23, 32]. Our assistant helps the patients navigate through the complicated procedure by sensing their actions and assisting intelligently, as shown in Figure 2.

3.2 LemurDx: Objective Measurement of Hyperactivity in Children

Hyperactivity is the predominant symptom of ADHD in children. Currently, it is assessed primarily through subjective reports from parents or teachers [33]. However, these reports can be inconsistent

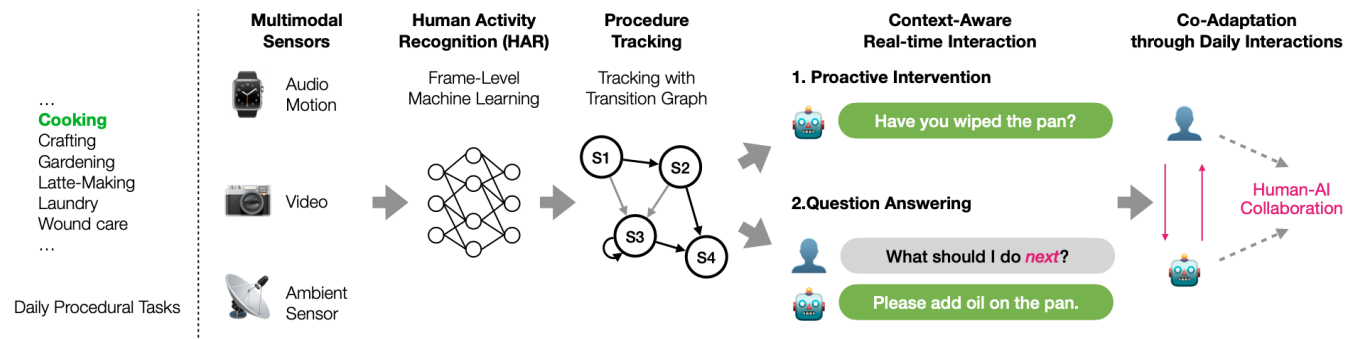


Figure 1: Thesis Project 1: *PrISM*—Procedural Interaction from Sensing Module—aims to support everyday procedural tasks (e.g., wound care for post-surgery skin cancer patient, daily tasks for people with dementia) using multimodal sensors (e.g., a smartwatch). Based on step-by-step tracking, it enables situated interactions such as reminders and question answering.

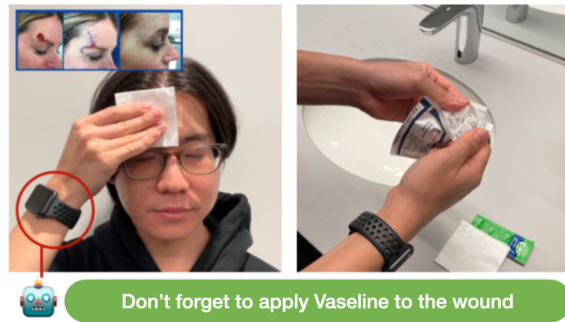


Figure 2: *PrISM* assistant can support post-operative skin cancer patients in a complicated wound care procedure.

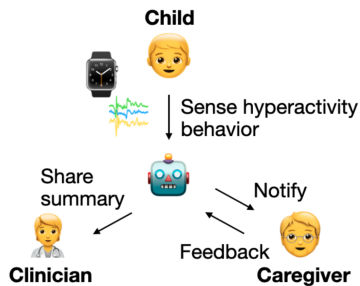


Figure 3: Thesis Project 2: *LemurDx* aims to measure children's hyperactivity from a smartwatch and assist surrounding people, such as caregivers and clinicians.

or biased, as caregivers are not always present, and their judgments may be influenced by context or perception. This subjectivity can lead to misdiagnoses and make it difficult to provide effective, personalized support. Thus, there is a need to add some objectivity to the diagnosis and measurement of hyperactivity.

To address this, I am developing a smartwatch-based assistant that objectively measures hyperactivity using its motion sensor and facilitates continuous communication among parents, caregivers, and clinicians, as illustrated in Figure 3. The assistant calculates a real-time risk score indicating the likelihood of hyperactive behavior in the child wearing the watch and sends timely notifications to caregivers. In response, caregivers can provide contextual information, such as the child's activity or their perceived level of hyperactivity, which helps refine the assistant's sensing model and contributes to a collaborative human-AI partnership. Additionally, these daily interactions generate rich behavioral data that can be shared with clinicians, enabling more tailored medical recommendations, such as adjusting medication dosages based on children's daily behavior and risk score patterns.

4 Evaluation

In this section, I first present the work already completed for each project described above, followed by the planned future work to be carried out as part of my thesis.

4.1 *PrISM* Project

4.1.1 Completed Work. I have developed multiple components for the pipeline shown in Figure 1. *PrISM-Tracker* (IMWUT2023) [19] introduces a step-tracking algorithm using multimodal HAR for procedural tasks. *PrISM-Observer* (UIST2024) [16] achieves a real-time reminder to prevent errors based on the tracking results. Similarly, *PrISM-Q&A* (IMWUT2024) [7] extends LLMs' question-answering capability with the step tracking to provide more contextually-situated answers to user questions.

In these studies, we have collected sensor data across various procedures involving post-operative skin cancer patients and individuals with dementia, in collaboration with the University Hospitals of Cleveland Department of Dermatology and Fraunhofer Portugal AICOS. Additionally, we conducted controlled in-lab studies to validate the effectiveness of *PrISM-Tracker*'s step-tracking algorithm, as well as the interactive components of *PrISM-Observer* and *PrISM-Q&A*. While these systems show promise, challenges remain, particularly when sensing errors lead to inappropriate or

mistimed interactions. To fully realize effective human-AI collaboration, further work is needed to enhance robustness and to evaluate these systems in real-world clinical contexts.

4.1.2 Future Work. I aim to deepen the interaction between the assistant and users to establish a collaborative relationship. Specifically, I will adopt a mixed-initiative design approach [3], a foundational principle in HCI for balancing user agency and system initiative. When the assistant detects signs of confusion or potential errors, it will proactively offer support. Conversely, when sensing is ambiguous or uncertain, the assistant will ask for user input, both clarifying context and enabling dynamic updates to its behavior models. This approach requires computational mechanisms that can extract actionable context from spontaneous, unstructured user interactions. I have implemented and evaluated such a mechanism [10], and will explore further techniques such as online reinforcement learning to leverage implicit user signals and continuously adapt the assistant’s understanding of behavior. Ultimately, such adaptive assistants will maintain user trust in the long term and continuously support their daily procedures, which is especially critical for patients with chronic conditions.

4.2 LemurDx Project

4.2.1 Completed Work. I developed a foundational pipeline for measuring children’s hyperactivity levels using a smartwatch, published in IMWUT2023 [4]. In collaboration with the University of Pittsburgh Medical Center, we collected data from 61 children (25 diagnosed with hyperactivity) who wore a smartwatch for up to 7 days without altering their daily routines. In such an unconstrained setting, simply quantifying children’s motion levels is not sufficient due to the lack of contextual understanding; for instance, children may exhibit high levels of movement during exercise regardless of their hyperactivity status. To address this, we developed models that estimate the child’s activity context using multimodal sensor data, which serves as prior information to assess hyperactivity severity based on motion levels. Our analysis demonstrated that this context-aware filtering approach effectively quantifies children’s hyperactivity. Nonetheless, the system’s accuracy is not perfect, underscoring the need for cautious interpretation and further refinement to support clinically valid outcomes.

4.2.2 Future Work. As one of the promising applications of the developed pipeline in [4], I plan to deploy a real-time monitoring assistant for parents or caregivers of children with hyperactivity. Based on the real-time hyperactivity measurement of LemurDx, this assistant will notify parents when specific patterns indicate a potential onset of heightened hyperactivity. Recognizing the challenges parents and caregivers face, such as predicting and managing hyperactive episodes and balancing these demands with daily responsibilities, this assistant will equip them with actionable insights derived from real-time data, enabling proactive management of hyperactivity. To improve the model’s accuracy through human-in-the-loop, we plan to introduce a feedback mechanism that enables parents and caregivers to provide contextual input about the child’s activities post-deployment. This feedback loop allows the assistant to continuously adapt to the unique dynamics of each child-caregiver relationship. By personalizing its responses and interventions over

time, the assistant aims to cultivate a more responsive and supportive environment that contributes to the child’s long-term growth and development.

5 Contribution

In my thesis, I will contribute to the following three themes:

- **Novel Sensing Technologies:** To enable context-aware health assistants, I have innovated in sensing technologies, including gaze tracking on smartphones [5], gesture recognition with a wrist-worn camera [9], 3D body pose estimation using commodity IMU-equipped devices [30, 34, 36] or electromagnetic field sensors [20], and the integration of machine-knitted scarves with electrical impedance tomography [35]. Most of these systems have been open-sourced to facilitate broader adoption and reproducibility. Furthermore, we are deploying HAR systems in the homes of target populations, such as individuals with dementia. The resulting multimodal dataset—capturing long-term, in-the-wild patient behaviors—will serve as a valuable resource for the broader research community in human activity recognition and behavior modeling.
- **Designs for Using Imperfect Sensing Reliably:** My work addresses the challenge of deploying sensing systems in real-world health contexts, where sensor data is often noisy or ambiguous. To achieve effective human-AI collaboration, I pursue sustainable adaptation techniques through interactions [6, 10, 17], design error-aware interventions [13, 18], and explore hybrid ways for incorporating domain experts’ insights into human behavior analysis [8, 11, 12, 14, 15]. I expect that these approaches, when combined, can sustain user trust even in the presence of imperfect sensing and modeling. My thesis will conclude with design strategies to use imperfect sensing reliably and pave the way for real-world human-AI collaboration.
- **Real-world Impact:** I have conducted multiple field deployments in collaboration with clinical partners. These efforts include supporting post-operative skin cancer patients in self-care tasks, people with dementia in daily activities, and quantifying hyperactivity in children using wearable sensing. The deployed systems are designed with input from medical experts and tested with patient populations to ensure relevance and feasibility. Beyond technical validation, these deployments aim to inform how intelligent, sensing-based assistants can be integrated into care workflows, bridging the gap between lab-based research and real-world clinical impact.

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